Lecture 11 - April 20

11.6 The Ratio and Root tests.

The Ratio test: Let consider a series $\sum_{n=1}^{\infty} a_n$.

i). If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L < 1$ then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (and therefore convergent).

ii). If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$ or $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \infty$ then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

iii). If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1$, the Ratio test is inconclusive.

Example: Test the series $\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{3^n}$

Sln: let $a_n = (-1)^n \frac{n^3}{3^n}$ then

$$\left|\frac{a_{n+1}}{a_n}\right| = \left| \frac{\frac{(-1)^{n+1}(n+1)^3}{3^{n+1}}}{\frac{(-1)^n n^3}{3^n}} \right| = \left| \frac{(-1)^{n+1}(n+1)^3 3^n}{(-1)^n 3^{n+1} n^3} \right|$$

$$= \left| \frac{(-1)(n+1)^3}{3 n^3} \right| = \frac{1}{3} \left(\frac{n+1}{n}\right)^3$$

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \frac{1}{3} \lim_{n\to\infty} \left(\frac{n+1}{n}\right)^3 = \frac{1}{3} < 1$ then $\sum_{n=1}^{\infty} \frac{(-1)^n n^3}{3^n}$ is absolutely convergent.

Example: Test the series $\sum_{n=1}^{\infty} \frac{n^n}{n!}$. ; let $a_n = \frac{n^n}{n!}$

Sln: $\left|\frac{a_{n+1}}{a_n}\right| = \left| \frac{(n+1)^{n+1} n!}{n^n (n+1)!} \right| = \frac{(n+1)^{n+1} \cancel{n!}}{n^n (n+1) \cancel{n!}} = \frac{(n+1)^n}{n^n} = \left(1 + \frac{1}{n}\right)^n$

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e > 1$ then the series $\sum_{n=1}^{\infty}{}' \frac{n^n}{n!}$ is divergent.

The root test: Let consider $\sum_1' a_n$

i) $\lim_{n\to\infty}\sqrt[n]{|a_n|} = L < 1$ then the series $\sum_{n=1}^{\infty}{}' a_n$ is absolutely convergent (and therefore convergent).

ii) If $\lim_{n\to\infty}\sqrt[n]{|a_n|} = L > 1$ or $\lim_{n\to\infty}\sqrt[n]{|a_n|} = \infty$ then $\sum_{n=1}^{\infty}{}' a_n$ is divergent

iii) If $\lim_{n\to\infty}\sqrt[n]{|a_n|} = 1$, the root test is inconclusive.

Example. Test the series $\sum_{n=1}^{\infty}{}' \left(\frac{2n+3}{3n+2}\right)^n$.

Sln: let $a_n = \left(\frac{2n+3}{3n+2}\right)^n$; $\sqrt[n]{|a_n|} = \sqrt[n]{\left(\frac{2n+3}{3n+2}\right)^n} = \left(\frac{2n+3}{3n+2}\right)$

So, $\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\frac{2n+3}{3n+2} = \frac{2}{3} < 1$ then the series $\sum_{n=1}^{\infty}{}' \left(\frac{2n+3}{3n+2}\right)^n$ is absolutely convergent.

Example: Test the series $\sum_{n=1}^{\infty}{}' \left(\frac{n+1}{n}\right)^n$

Sln. let $a_n = \left(\frac{n+1}{n}\right)^n$ then

$$\sqrt[n]{|a_n|} = \sqrt[n]{\left(\frac{n+1}{n}\right)^n} = \left(\frac{n+1}{n}\right)$$

$$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\left(\frac{n+1}{n}\right) = 1.$$

$\lim_{n\to\infty}\left(\frac{n+1}{n}\right)^n = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e \neq 0$ then By the divergent test $\sum_{n=1}^{\infty}{}' \left(\frac{n+1}{n}\right)^n$ is divergent.

11.8 Power Series. So far we have studied series of numbers $\sum a_n$. Here, we consider power series, in which each term includes a power of the variable x: $\sum_{n=0}^{\infty} a_n \cdot x^n$

Rmk: i) $\sum_{n=1}^{\infty} x^n$ $\quad a_n = 1$ for all n.

$$= x + x^2 + x^3 + \cdots + x^n + \cdots$$

if $x = \frac{1}{2} \Rightarrow \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ is convergent!

if $x = -\frac{1}{2} \Rightarrow \sum_{n=1}^{\infty} \left(-\frac{1}{2}\right)^n$ is absolutely convergent!

if $x = 1 \Rightarrow \sum_{n=1}^{\infty} 1^n$ is divergent.

then $f(x) = \sum_{n=1}^{\infty} x^n$ for $|x| < 1$.

Example: For what values of x does the series $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ converge?

Sln: let $a_n = \frac{(x-3)^n}{n}$.

$$\left|\frac{a_{n+1}}{a_n}\right| = \left| \frac{(x-3)^{n+1}}{n+1} \cdot \frac{n}{(x-3)^n} \right| = |x-3| \left(\frac{n}{n+1}\right)$$

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} |x-3| \left(\frac{n}{n+1}\right)$$

$$= |x-3| \lim_{n\to\infty} \frac{n}{n+1}$$

$$= |x-3| < 1 \quad \text{(condition of the Ratio test)}.$$

Solving $|x-3| < 1$: $\{ |y| < a \iff -a < y < a \}$.

$$-1 < x-3 < 1$$

$$2 < x < 4.$$

Now for the end points: If $x=2 \Rightarrow \sum_{n=1}^{\infty} \frac{(2-3)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$.

this series is convergent by the A-t. b/c:

let $b_n = \frac{1}{n}$, $\lim_{n\to\infty} \frac{1}{n} = 0$ and $n+1 > n \Rightarrow \frac{1}{n} = b_n > \frac{1}{n+1} = b_{n+1}$

so $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent.

If $x=4$: $\Rightarrow \sum_{n=1}^{\infty} \left(\frac{4-3}{n}\right)^n = \sum_{n=1}^{\infty} \frac{1}{n}$ the harmonic series which is divergent.

Therefore, $f(x) = \sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ where $\text{Dom } f = [2,4)$.

Example: For what values of x is the series $\sum_{n=0}^{\infty} n!\, x^n$ convergent?

Sln: